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Inventor(s)

Tsang; Albert et al.

CABLE CONNECTOR ASSEMBLY HAVING CONNECTOR IDENTIFICATION MODULE

Abstract

A cable connector assembly includes a cable connector including a connector housing holding contact assemblies with cables terminated to signal contacts and shields. The cable connector assembly includes a connector identification module configured to be mated with a mating connector assembly. The connector identification module includes an identification contact assembly received in the cavity and coupled to the connector housing. The identification contact assembly includes an identification contact holder holding an identification signal contact. The connector identification module includes an identification device having unique identification data stored on the identification device. The identification device is operably coupled to the identification signal contact to transmit the unique identification data to the mating connector assembly.

Inventors: Tsang; Albert (Middletown, PA), Lachman; Julia Anne (Middletown, PA), Trout; David Allison (Middletown, PA), Shields; Linda Ellen (Middletown, PA), Hancock; Jazmine (Middletown, PA)

Applicant: TE Connectivity Solutions GmbH (Schaffhausen, CH)

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application claims benefit to U.S. Provisional Application No. 63/563,511, filed 11 Mar. 2024, titled “CABLE CONNECTOR ASSEMBLY HAVING CONNECTOR IDENTIFICATION MODULE”, the subject matter of which is herein incorporated by reference in its entirety.

BACKGROUND OF THE INVENTION

[0002] The subject matter herein relates generally to communication systems.

[0003] Cable assemblies are used to electrically connect various components within a system, such as a server or network system. The cable assemblies extend between the components, such as network, compute, storage, memory, or other components. The components may be arranged in shelves or trays of a network rack. Some systems may include components such as cards, such as line cards, daughter cards, mother boards, and the like. Each cable assembly includes multiple cables arranged in a bundle or array. Each cable forms a data channel. As the density of cable assemblies in the network system increase, it is desirable to know the locations of the components within the system, such as for service, repair or replacement.

[0004] A need remains for a method and device for identifying components and locations of components within a network system.

BRIEF DESCRIPTION OF THE INVENTION

[0005] In one embodiment, a cable connector assembly is provided and includes a cable connector including a connector housing having walls to define a cavity. The cable connector includes contact assemblies received in the cavity. Each contact assembly includes a contact holder holding a signal contact and a shield providing shielding for the signal contact. The signal contacts and the shields of the contact assemblies arranged at a mating end of the cable connector configured to be mated with a mating connector assembly. The cable connector assembly includes cables terminated to the signal contacts and extend from a cable end of the cable connector. The cables define data channels. The cable connector assembly includes a connector identification module configured to be mated with the mating connector assembly. The connector identification module includes an identification contact assembly received in the cavity and coupled to the connector housing. The identification contact assembly includes an identification contact holder holding an identification signal contact arranged at the mating end of the cable connector configured to be mated with the mating connector assembly. The connector identification module includes an identification device having unique identification data stored on the identification device. The identification device is operably coupled to the identification signal contact to transmit the unique identification data to the mating connector assembly.

[0006] In another embodiment, a connector identification module is provided and includes an identification contact assembly including an identification contact holder extending between a front and a rear. The identification contact assembly includes an identification signal contact held by the identification contact holder. The identification signal contact includes a mating end and a terminating end. The mating end configured to be mated with a mating connector assembly. The cable connector assembly includes an identification cable extending between a first end and a second end. The first end terminated to the identification signal contact. The cable connector assembly includes an identification device having a printed circuit board assembly that includes a printed circuit board and an electronic device mounted to the printed circuit. The electronic device

has unique identification data stored on the electronic device. The printed circuit board is terminated to the second end of the identification cable. The cable defines a data channel between the printed circuit board and the identification signal contact to transmit the unique identification data to the mating connector assembly.

[0007] In a further embodiment, a communication system is provided and includes a cartridge having panels forming a chamber. The cartridge includes ports at a front of the cartridge. The communication system includes cable connector assemblies received in the ports for mating with mating connector assemblies. Each cable connector assembly includes a cable connector and a connector identification module communicatively coupled to the cable connector by cables. Each cable connector includes a connector housing having walls define a cavity. The cable connector includes contact assemblies received in the cavity. Each contact assembly includes a contact holder holding a signal contact and a shield providing shielding for the signal contact. The signal contacts and the shields of the contact assemblies arranged at a mating end of the cable connector configured to be mated with the mating connector assembly. The cables are terminated to the signal contacts and extend from a cable end of the cable connector. The cables define data channels. The connector identification module includes an identification contact assembly received in the cavity and coupled to the connector housing. The identification contact assembly includes an identification contact holder holding an identification signal contacts arranged at the mating end of the cable connector configured to be mated with the mating connector assembly. The connector identification module includes an identification device having unique identification data stored on the identification device. The identification device is operably coupled to the identification signal contacts to transmit the unique identification data to the mating connector assembly.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is a front view of a communication system in accordance with an exemplary embodiment.

[0009] FIG. 2 is a front view of a portion of the communication system in accordance with an exemplary embodiment.

[0010] FIG. 3 illustrates a portion of the communication system showing two of the cartridges in accordance with an exemplary embodiment.

[0011] FIG. 4 is a perspective view of a portion of the communication system showing one of the mating connector assemblies poised for coupling to the corresponding connector assembly in accordance with an exemplary embodiment.

[0012] FIG. 5 is a front perspective view of a portion of the communication system\showing one of the connector assemblies\in accordance with an exemplary embodiment.

[0013] FIG. 6 is an exploded view of one of the contact assemblies in accordance with an exemplary embodiment.

[0014] FIG. 7 is a side view of one of the contact modules in accordance with an exemplary embodiment.

[0015] FIG. 8 is a side view of a portion of the connector identification module showing an identification contact module in accordance with an exemplary embodiment.

[0016] FIG. 9 is a side view of a portion of the connector identification module showing the identification device in accordance with an exemplary embodiment.

[0017] FIG. 10 illustrates one of the connector assemblies in accordance with an exemplary embodiment having the connector identification module.

[0018] FIG. 11 illustrates a portion of the communication system showing a plurality of the connector assemblies in accordance with an exemplary embodiment arranged in the cartridge.

DETAILED DESCRIPTION OF THE INVENTION

[0019] FIG. 1 is a front view of a communication system **100** in accordance with an exemplary embodiment. FIG. 2 is a front view of a portion of the communication system **100** in accordance with an exemplary embodiment. In an exemplary embodiment, the communication system **100** includes a chassis or rack **50** having a rack frame **52** holding electrical components **60**. The rack **50** may be a server rack. The electrical components **60** may include fixed modules **62** (FIG. 2) and/or removable modules **64** (FIG. 1). The removable modules **64** may be mated to the fixed modules **62**. For example, the removable modules **64** may be plugged into the rack frame **52** and removable from the rack frame **52**. In various embodiments, the electrical components **60** may include one or more patch panels, switches, servers, routers, firewalls, and the like. The rack frame **52** may hold a power supply, cooling components, and the like.

[0020] In an exemplary embodiment, the communication system **100** includes at least one connector assembly **102** (FIG. 2) and at least one mating connector assembly **104** (FIG. 4) configured to be mated with the corresponding at least one connector assembly **102**. The connector assembly **102** may be part of one of the electrical components **60**, such as one of the fixed modules **62**. The mating connector assembly **104** may be part of one of the electrical components **60**, such as one of the removable modules **64**. In various embodiments, each mating connector assembly **104** may include one or more mating electrical connectors mounted to a circuit board, such as a backplane, a daughter card, a network switch, and the like.

[0021] In an exemplary embodiment, the communication system **100** includes a connector identification system **200** to identify the various connector assemblies **102** and/or mating connector assemblies **104**. The connector identification system **200** may be used to identify physical locations of the connector assemblies **102** within the communication system **100**, such as to identify which connector in a cable harness is electrically connected to which electrical component **60** for system management. In an exemplary embodiment, the connector identification system **200** may use identification devices, such as EEPROMs (Electrically Erasable and Programmable Read Only Memory), to hold information that allows identification of the various connector assemblies **102**. The identification devices may be incorporated with each of the connector assemblies **102**. The identification devices may have unique identification information programmed into the identification devices, such as a part number, a program name, a slot ID number, and the like. In an exemplary embodiment, the identification devices are modular and may be incorporated into different types of connector assemblies **102**.

[0022] In an exemplary embodiment, the communication system **100** includes a cabinet or cartridge **110** that holds the connector assemblies **102**. The connector assemblies **102** may be cable connector assemblies having cables extending from corresponding cable connectors. The cartridge **110** may form part of the fixed modules **62**. Optionally, multiple cartridges **110** may be provided (for example, left side and right side). The cartridge **110** may be coupled to the rack frame **52**, such as at the rear of the rack frame **52**. The cartridge **110** forms an enclosed space or chamber **111** for the connector assemblies **102** and the cables of the connector assemblies **102**. The cartridge **110** is used for cable management. For example, the cables may be part of cable harnesses routed within the enclosed chamber **111** formed by the cartridge **110** electrically connecting the various connector assemblies **102**. The cartridge **110** is used to support or position the connector assemblies **102** for mating with the mating connector assemblies **104**.

[0023] FIG. 3 illustrates a portion of the communication system **100** showing two of the cartridges **110** in accordance with an exemplary embodiment. Each cartridge **110** holds a plurality of the connector assemblies **102**, such as in one or more rows and one or more columns. The connector assemblies **102** are provided at the front of the cartridge **110** for interfacing with the mating connector assemblies **104** (shown in FIG. 4).

[0024] In an exemplary embodiment, the cartridge **110** is formed from a plurality of panels **112**, such as sheet metal panels. The cartridge **110** includes a front panel **114**, side panels **116**, and a rear

panel **118**. The panels **112** may additionally include an upper panel and/or a lower panel. The connector assemblies **102** are provided at the front panel **114** for mating with the mating connector assemblies **104**. In an exemplary embodiment, the connector assemblies **102** are arranged in a column along the front panel **114**. Optionally, the connector assemblies **102** may be provided in multiple columns. The mating ends of the connector assemblies **102** are external to the cartridge **110**. For example, the connector assemblies **102** pass through ports or openings **120** in the front panel **114**. The connector assemblies **102** are coupled to the front panel **114** adjacent the openings **120**. In an exemplary embodiment, the connector assemblies **102** may float or move relative to the front panel **114** to align with and mate to the mating connector assemblies **104**. In an exemplary embodiment, the connector assemblies **102** include mating guides, such as guide posts **122**, to guide mating of the mating electrical connectors **106** with the connector assemblies **102**. The guide posts **122** extend forward of the front panel **114**.

[0025] FIG. **4** is a perspective view of a portion of the communication system **100** showing one of the mating connector assemblies **104** poised for coupling to the corresponding connector assembly **102**. The connector assembly **102** is coupled to the cartridge **110**, such as to the front panel **114**. In the illustrated embodiment, the mating connector assembly **104** is a network component. For example, the mating connector assembly **104** may be a compute node, a memory module, a network switch, a storage device, a network accelerator, or another type of communication component. The mating connector assembly **104** may be a backplane component or a daughtercard component.

[0026] In an exemplary embodiment, the mating connector assembly **104** includes a mating electrical connector **106** mounted to a circuit board **108**. The mating connector assembly **104** may be mounted to a front edge of the circuit board **108**. The mating electrical connector **106** includes a connector housing **109** holding signal contacts (not shown). The signal contacts are terminated to the circuit board **108**. The signal contacts may be socket contacts. The mating electrical connector **106** may include shields providing shielding for the signal contacts.

[0027] The connector assembly **102** is received in the cartridge **110**. The connector assembly **102** is provided at the front panel **114** for mating with the mating connector assembly **104** along a mating axis **126**. The mating end of the cable connector **130** is configured to pass through an opening in the front panel **114** for mating with the mating electrical connector **106**. The connector assemblies **102**, **104** are configured to transmit and/or receive data through an interface. The connector assemblies **102** may include receptacle connectors and the mating connector assemblies **104** may include plug connectors. In an exemplary embodiment, the guide post **122** is configured to be mated with a guide module **124** of the mating connector assembly **104** to guide mating of the mating connector assembly **104** with the connector assembly **102**. In the illustrated embodiment, the guide module **124** includes an opening that receives the guide post **122**. The mating guides may be used to align the connectors in one or more lateral directions.

[0028] FIG. **5** is a front perspective view of one of the connector assemblies **102** in accordance with an exemplary embodiment. In an exemplary embodiment, the connector assembly **102** includes a cable connector **130** having a connector identification module **202**. The connector identification module **202** is part of the connector identification system **200** used to identify the connector assembly **102** within the system. In an exemplary embodiment, the connector identification module **202** includes one or more identification contact assemblies **240** and an identification device **210** operably coupled to the identification contact assemblies **240**. The identification contact assemblies **240** may be arranged as parts of an identification contact module **260**. In an exemplary embodiment, the identification device **210** is connected to the identification contact assemblies **240** by an identification cable **250**. For example, the identification device **210** is an EEPROM dongle hanging from the identification contact module **260**.

[0029] The cable connector **130** includes a connector housing **132** holding the connector identification module **202** and a plurality of contact modules **160**. The contact modules **160** include

contact assemblies **140** configured to be mated with the mating connector assembly **104**. The contact assemblies **140** are terminated to ends of the cables **150**. The contact assemblies **140** are arranged together within the contact modules **160**. For example, the contact modules **160** hold the contact assemblies **140** in columns. A plurality of the contact modules **160** are received in the connector housing **132**, such as being arranged in a contact module stack. In an exemplary embodiment, the connector identification module **202** is arranged in the contact module stack, such as being arranged at one of the ends of the contact module stack. The cables **150** extend from the rear ends of the contact modules **160**.

[0030] The connector housing **132** includes a cavity **134** that receives the contact modules **160** and the connector identification module **202**. The contact assemblies **140** are arranged in the cavity **134** at a mating end for mating with the mating electrical connector **106**. For example, the contact assemblies **140** are arranged in the cavity **134** in rows and columns. The cavity **134** may form a receptacle that receives the mating end of the mating electrical connector **106**. The walls of the connector housing **132** may be chamfered and have a lead-in surfaces to guide mating of the mating electrical connector **106** in the cavity **134**. The connector housing **132** may have guide features to properly position the mating electrical connector **106** within the cavity **134**.

[0031] With additional reference to FIG. **6**, which is an exploded view of one of the contact assemblies **140**, each contact assembly **140** includes at least one signal contact **142**, a contact holder **141** holding the at least one signal contact **142**, and a shield **144** coupled to the contact holder **141** and providing electrical shielding for the at least one signal contact **142**.

[0032] In the illustrated embodiment, the contact assembly **140** includes a pair of the signal contacts **142**, which define a differential pair. In various embodiments, the signal contacts **142** may be pin contacts. However, in alternative embodiments, the signal contacts **142** may be socket contacts, spring beam contacts, or other types of contacts. The signal contacts **142** may be stamped and formed contacts. Each signal contact **142** is configured to be electrically connected to a corresponding wire or conductor of the cable **150**. In an exemplary embodiment, the cable **150** is a twin-axial cable having a first conductor and a second conductor terminated to corresponding signal contacts **142**.

[0033] The contact holder **141** is manufactured from a dielectric material, such as a plastic material. The contact holder **141** may be molded. The contact holder **141** includes a contact channel **143** for each signal contact **142**. In the illustrated embodiment, the contact holder **141** has two contact channels **143** for the pair of signal contacts **142**. In various embodiments, the contact holder **141** is preformed and the signal contacts **142** are loaded into the contact holder **141**. In other various embodiments, the contact holder **141** is formed in situ around the signal contacts **142**, such as being overmolded over the signal contacts **142**. The cable **150** may extend into an end of the contact holder **141** to terminate to the signal contacts **142**. The contact holder **141** may provide strain relief for the end of the cable **150**.

[0034] The shield **144** may be a stamped and formed shield. The shield **144** is configured to be coupled to the contact holder **141**. The shield **144** may be a multi-piece shield having an upper shield **146** and a lower shield **148**. The upper and lower shields **146**, **148** are coupled together to provide circumferential shielding for the end of the cable **150** and the signal contacts **142**. The shield **144** is configured to be electrically connected to the cable shield of the cable. In an exemplary embodiment, the front end of the shield **144** is C-shaped providing electrical shielding on three sides of the pair of signal contacts **142**. The shield **144** may have other shapes in alternative embodiments.

[0035] FIG. **7** is a side view of one of the contact modules **160** in accordance with an exemplary embodiment. The contact module **160** includes a body **162** holding the contact assemblies **140**. The cables **150** extend from a cable end of the contact module **160**. The signal contacts **142** and the shields **144** extend from a mating end of the contact module **160**. In an exemplary embodiment, the body **162** holds a ground shield **164** configured to be electrically connected to the shields **144**. The

body **162** includes latches **166**, such as at the top and the bottom of the body **162**, used to secure the contact module **160** in the connector housing **132** (shown in FIG. 5). The contact module **160** may include a strain relief **168**, such as at the cable end, configured to hold the ends of the cables **150** and provide strain relief for the cables **150**. The strain relief **168** may be an overmold overmolded over the ends of the cables **150**.

[0036] FIG. 8 is a side view of a portion of the connector identification module **202** showing an identification contact module **260** in accordance with an exemplary embodiment. The identification contact module **260** may be similar to the signal contact module **160** (shown in FIG. 7). For example, the identification contact module **260** and the contact module **160** may have similar dimensions and components. The identification contact module **260** and the contact module **160** may be interchangeable within the contact module stack and arranged in the connector housing **132** for mating with the mating connector assembly **104**. In an exemplary embodiment, the identification contact module **260** includes a plurality of the contact assemblies **240** configured to be mated with the mating connector assembly **104**.

[0037] The identification contact module **260** includes a body **262** holding the identification contact assemblies **240**. The identification cables **250** extend from a cable end of the identification contact module **260**. The identification contact assemblies **240** may be similar to the contact assembly **140** (shown in FIG. 6). Each identification contact assembly **240** includes at least one identification signal contact **242**, a contact holder **241** holding the at least one identification signal contact **242** in contact channels, and a shield **244** coupled to the contact holder **241** and providing electrical shielding for the at least one identification signal contact **242**. In an exemplary embodiment, the identification contact assembly **240** includes a pair of the identification signal contacts **242**, which define a differential pair. Each identification signal contact **242** is configured to be electrically connected to a corresponding wire or conductor of the identification cable **250**. In an exemplary embodiment, the identification cable **250** is a twin-axial cable having a first conductor and a second conductor terminated to corresponding identification signal contacts **242**. The shield **244** is configured to be electrically connected to the cable shield of the cable. In an exemplary embodiment, the front end of the shield **244** is C-shaped providing electrical shielding on three sides of the pair of signal contacts **242**.

[0038] The identification signal contacts **242** and the shields **244** extend from a mating end of the identification contact module **260**. In an exemplary embodiment, the body **262** holds a ground shield **264** configured to be electrically connected to the shields **244**. The body **262** includes latches **266**, such as at the top and the bottom of the body **262**, used to secure the identification contact module **260** in the connector housing **132** (shown in FIG. 5). The identification contact module **260** may include a strain relief **268**, such as at the cable end, configured to hold the ends of the identification cables **250** and provide strain relief for the identification cables **250**. The strain relief **268** may be an overmold overmolded over the ends of the identification cables **250**. In the illustrated embodiment, the identification contact module **260** includes three of the identification contact assemblies **240**, which may correspond to hot, ground, and neutral for the connector identification module **202**.

[0039] FIG. 9 is a side view of a portion of the connector identification module **202** showing the identification device **210** in accordance with an exemplary embodiment. The identification device **210** is provided at ends of the identification cables **250** opposite the identification contact module **260** (shown in FIG. 8). The identification device **210** is used to identify the connector assembly **102** associated with the connector identification module **202**. The identification device **210** may be used to identify physical locations of the connector assembly **102** within the communication system **100**. In an exemplary embodiment, the identification device **210** may be an EEPROM. The identification device **210** holds information that allows identification of the connector assembly **102**. The identification device **210** may have unique identification information programmed into the identification device **210**, such as a part number, a program name, a slot ID number, and the

like. The identification device **210** is operably coupled to the identification signal contacts **244** to transmit the unique identification data to the mating connector assembly **104**.

[0040] In an exemplary embodiment, the identification device **210** includes an electronic device **212** that stores unique identification data. The electronic device **212** may be an integrated circuit, such as a chip. In an exemplary embodiment, the electronic device **212** is an EEPROM chip. The electronic device **212** is programmable to store the identification data, such as a part number, a program name, a slot ID number, and the like.

[0041] In an exemplary embodiment, the identification device **210** includes a printed circuit board assembly **220** including a printed circuit board **222**, the electronic device **212** mounted to the printed circuit board **222** and may include other components mounted to the printed circuit board **222**. The printed circuit board **222** includes pads **224** on one or more surfaces of the printed circuit board **222**. The identification cables **250** are terminated to the pads **224**, such as being soldered to the pads **224**.

[0042] In an exemplary embodiment, the identification device **210** includes a device shell **230** holding the printed circuit board assembly **220**. The device shell **230** surrounds the printed circuit board **222** and the electronic device **212**. In an exemplary embodiment, the device shell **230** includes an overmold body **232** overmolded over the printed circuit board assembly **220**. In alternative embodiments, the device shell **230** may include shells coupled together to hold the printed circuit board assembly **220** therebetween. The device shell **230** surrounds the printed circuit board assembly **220** to protect the components, such as the electronic device **212**. In an exemplary embodiment, the device shell **230** holds the identification cables **250**. For example, the overmold body **232** may be overmolded over the ends of the identification cables **250**. The device shell **230** may provide strain relief for the identification cables **250**.

[0043] FIG. **10** illustrates one of the connector assemblies **102** in accordance with an exemplary embodiment having the connector identification module **202**. FIG. **11** illustrates a portion of the communication system **100** showing a plurality of the connector assemblies **102** in accordance with an exemplary embodiment arranged in the cartridge **110**. Some of the panels **112** of the cartridge **110** are removed to illustrate the connector assemblies **102**, the cables **150** between the connector assemblies **102**, and the connector identification modules **202** associated with each of the connector assemblies **102**. The cables **150** are part of cable harnesses **152** that connect the various connector assemblies **102**. The cable harnesses **152** are contained in the cartridge **110**, such as in the enclosed chamber **111** defined by the panels **112**.

[0044] When assembled, the connector identification module **202** and the contact modules **160** are loaded into the connector housing **132**. For example, the connector identification module **202** and the contact modules **160** are arranged in the cavity **134**. The connector identification module **202** and the contact modules **160** may be received in contact assembly slots **136** in the cavity **134**. The connector identification module **202** and the contact modules **160** may be interchangeably receivable in the contact assembly slots **136**. For example, because the connector identification module **202** and the contact modules **160** have the same dimensions and mating features, the connector identification module **202** may be arranged in any of the slots.

[0045] The cables **150** extend from the cable side of the cable connector **130**, such as to another cable connector **130**. The identification cables **250** extend from the cable side of the cable connector **130** to allow the identification device **210** to dangle or hang from the cable connector **130**. For example, the identification device **210** may be an EEPROM dongle hanging from the cable connector **130**. The identification device **210** is remote from the connector housing **132**. For example, the identification cables **250** have sufficient length to allow the identification device **210** to hang in the enclosed chamber **111** spaced apart from the connector housing **132**. In an exemplary embodiment, a first end **252** of each identification cable **250** is terminated to the identification signal contact **242** of the identification contact assembly **240**. A second end **254** of each identification cable **250** is terminated to the identification device **210**, such as to the printed circuit

board 222. The identification cables 250 define data channels between the electronic device 212 on the printed circuit board 222 and the identification signal contacts 244 to transmit the unique identification data to the mating connector assembly.

[0046] It is to be understood that the above description is intended to be illustrative, and not restrictive. For example, the above-described embodiments (and/or aspects thereof) may be used in combination with each other. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the invention without departing from its scope.

Dimensions, types of materials, orientations of the various components, and the number and positions of the various components described herein are intended to define parameters of certain embodiments, and are by no means limiting and are merely exemplary embodiments. Many other embodiments and modifications within the spirit and scope of the claims will be apparent to those of skill in the art upon reviewing the above description. The scope of the invention should, therefore, be determined with reference to the appended claims, along with the full scope of equivalents to which such claims are entitled. In the appended claims, the terms “including” and “in which” are used as the plain-English equivalents of the respective terms “comprising” and “wherein.” Moreover, in the following claims, the terms “first,” “second,” and “third,” etc. are used merely as labels, and are not intended to impose numerical requirements on their objects. Further, the limitations of the following claims are not written in means-plus-function format and are not intended to be interpreted based on 35 U.S.C. § 112(f), unless and until such claim limitations expressly use the phrase “means for” followed by a statement of function void of further structure.

Claims

1. A cable connector assembly comprising: a cable connector including a connector housing having walls defining a cavity, the cable connector including contact assemblies received in the cavity, each contact assembly including a contact holder holding a signal contact and a shield providing shielding for the signal contact, the signal contacts and the shields of the contact assemblies arranged at a mating end of the cable connector configured to be mated with a mating connector assembly; cables terminated to the signal contacts and extending from a cable end of the cable connector, the cables defining data channels; and a connector identification module configured to be mated with the mating connector assembly, the connector identification module including an identification contact assembly received in the cavity and coupled to the connector housing, the identification contact assembly including an identification contact holder holding an identification signal contact arranged at the mating end of the cable connector configured to be mated with the mating connector assembly, the connector identification module including an identification device having unique identification data stored on the identification device, the identification device being operably coupled to the identification signal contact to transmit the unique identification data to the mating connector assembly.

2. The cable connector assembly of claim 1, wherein the connector identification module includes an identification cable between the identification contact assembly and the identification device.

3. The cable connector assembly of claim 1, wherein the identification device is remote from the connector housing, the identification device being connected to the identification contact assembly by an identification cable.

4. The cable connector assembly of claim 1, wherein the connector identification module includes an identification cable extending between a first end and a second end, the first end being terminated to the identification signal contact, the second end being terminated to the identification device.

5. The cable connector assembly of claim 5, wherein the identification device includes a printed circuit board assembly including a printed circuit board and an electronic device mounted to the printed circuit board, the unique identification data being stored on the electronic device, wherein

the second end of the identification cable is terminated to the printed circuit board, the identification cable defining a data channel between the printed circuit board and the identification signal contact to transmit the unique identification data to the mating connector assembly.

6. The cable connector assembly of claim 5, wherein the identification device is an EEPROM dongle hanging from the identification contact assembly.

7. The cable connector assembly of claim 5, wherein the identification device includes a device shell surrounding the printed circuit board assembly.

8. The cable connector assembly of claim 7, wherein the device shell includes an overmold body overmolded over the printed circuit board, the electronic device, and the second end of the identification cable.

9. The cable connector assembly of claim 5, wherein the electronic device includes an EEPROM chip.

10. The cable connector assembly of claim 5, wherein the identification cable is a first identification cable, the connector identification module further comprising a second identification cable terminated to the printed circuit board and a second identification signal contact and a third identification cable terminated to the printed circuit board and a third identification signal contact.

11. The cable connector assembly of claim 5, wherein the identification cable is a twinaxial cable having a first conductor and a second conductor, the first conductor terminated to the identification signal contact, the second conductor terminated to a second identification signal contact of the identification contact assembly.

12. The cable connector assembly of claim 5, wherein the identification contact assembly includes a shield providing shielding for the identification signal contact, the shield being electrically connected to a cable shield of the identification cable.

13. The cable connector assembly of claim 1, wherein the connector housing includes contact assembly slots in the cavity, the contact assemblies and the identification contact assembly being interchangeably receivable in the contact assembly slots.

14. The cable connector assembly of claim 1, wherein the identification contact assembly is sized and shaped the same as the contact assemblies of the cable connector.

15. The cable connector assembly of claim 1, wherein each contact assembly includes latches being latchably coupled to the connector housing to secure the contact assembly in the cavity, the identification contact assembly including latches being latchably coupled to the connector housing to secure the identification contact assembly in the cavity.

16. A connector identification module comprising: an identification contact assembly includes an identification contact holder extending between a front and a rear, the identification contact assembly including an identification signal contact held by the identification contact holder, the identification signal contact including a mating end and a terminating end, the mating end configured to be mated with a mating connector assembly; an identification cable extending between a first end and a second end, the first end terminated to the identification signal contact; and an identification device having a printed circuit board assembly including a printed circuit board and an electronic device mounted to the printed circuit, the electronic device having unique identification data stored on the electronic device, the printed circuit board being terminated to the second end of the identification cable, wherein the cable defines a data channel between the printed circuit board and the identification signal contact to transmit the unique identification data to the mating connector assembly.

17. The connector identification module of claim 16, wherein the identification device is an EEPROM dongle hanging from the identification contact assembly.

18. The connector identification module of claim 16, wherein the identification device includes a device shell surrounding the printed circuit board assembly.

19. A communication system comprising: a cartridge having panels forming a chamber, the cartridge including ports at a front of the cartridge; cable connector assemblies received in the ports

for mating with mating connector assemblies, each cable connector assembly including a cable connector and a connector identification module communicatively coupled to the cable connector by cables, each cable connector including a connector housing having walls defining a cavity, the cable connector including contact assemblies received in the cavity, each contact assembly including a contact holder holding a signal contact and a shield providing shielding for the signal contact, the signal contacts and the shields of the contact assemblies arranged at a mating end of the cable connector configured to be mated with the mating connector assembly; the cables terminated to the signal contacts and extending from a cable end of the cable connector, the cables defining data channels; and the connector identification module including an identification contact assembly received in the cavity and coupled to the connector housing, the identification contact assembly including an identification contact holder holding an identification signal contacts arranged at the mating end of the cable connector configured to be mated with the mating connector assembly, the connector identification module including an identification device having unique identification data stored on the identification device, the identification device being operably coupled to the identification signal contacts to transmit the unique identification data to the mating connector assembly.

20. The communication system of claim 19, wherein the connector identification modules uniquely identify the positions of the corresponding cable connector assemblies within the communication system based on the unique identification data.
